



Electrochemical performance of a DNA-based sensor device for detecting toxic algae

Jahir Orozco^{a,b,*}, Linda K. Medlin^{a,b}

^a UPMC Univ Paris 06, UMR 7621, LOMIC, Observatoire Océanologique, F-66651 Banyuls/mer, France

^b CNRS, UMR 7621, LOMIC, Observatoire Océanologique, F-66651 Banyuls/mer, France

ARTICLE INFO

Article history:

Received 1 July 2010

Received in revised form

30 September 2010

Accepted 8 October 2010

Available online 16 October 2010

Keywords:

DNA-sensor

Sandwich hybridization

Toxic algae

Screen printed electrode

Prymnesium

Environmental monitoring

ABSTRACT

This work illustrates the electrochemical performance of a DNA-based sensor device for detecting toxic algae. This biosensor uses an electrochemical detection of the species in a sandwich hybridisation. A thiol (biotin) labelled capture probe was immobilized onto gold (carbon) electrodes. Synthetic positive control DNA was applied to the sensor and allowed to hybridize to the capture probe. A signal probe with a horseradish peroxidase (HRP) label was then applied, followed by an antibody to the HRP and a substrate. The electrical signal obtained from the redox reaction was proportional to the amount of DNA applied to the biosensor, which in turn would be proportional to the number of cells harvested when applied to real samples. Optimization of the hybridization process was already achieved in a previous work. Elucidation of the different steps of the fabrication process from the electrochemical point of view, proof of concept with different algal species and evaluation of the influence of the transducer platform geometry and material in the biosensor analytical performance are the main achievements reported here.

© 2010 Elsevier B.V. All rights reserved.

1. Introduction

The increase in blooms of microalgae, specially those ca. 85 species considered to be toxic [1], can be potentially harmful for shellfish (e.g., oysters, mussels) and fish, thus producing direct or indirectly extensive economic damage to fisheries, aquaculture and tourism and even causing public health problems, i.e., toxic algae can produce amnesic, paralytic or diarrhetic shellfish poisoning [2–4]. In this context, monitoring of water along coastlines for toxin-producing plankton and toxins in mussels is a requirement of the European Union (Directive 91/492d/EC and Commission Decision 2004/41/EC). In the past, monitoring and mitigation efforts have relied on visual confirmation of water discoloration, fish kills, and laborious cell counts. These practices are very time consuming, require specialized or trained personnel, expensive equipment, and then they are inappropriate when many samples have to be routinely analysed. Conventional methods for species detection often require transportation of samples to specialized laboratories, and identification of species can take several days. When results are obtained within five working days after receiving the sample, preventive measures to reduce the impact of harmful

algal blooms (HABs) are not always possible. Among molecular methods, which are potentially faster and more accurate than conventional ones, those based on sandwich hybridization assays have also been used for identification of phytoplankton, not just those that are toxic [5–12]. However, features of sensors and their benefits make them promising analytical tools to monitor and predict bloom formation and to mitigate their harmful effects on the local communities and surrounding ecosystems. Preliminary data on an innovative approach to HABs cell enumeration based on fibre optic genosensors for *Alexandrium fundyense*, *Pseudo-nitzschia australis* and *Alexandrium ostenfeldii* have been introduced [13]. A faster, more cheap, friendly use and reliable biosensor method, together with a handheld device for the detection and identification of the toxic dinoflagellate *Alexandrium ostenfeldii* and *A. minutum* have been reported [14]. Development and adaptation of a multiprobe biosensor for the use in a semi-automated device for the simultaneous detection of 14 target species of toxic algae have been also recently devised [15]. However, a multianalyte system not only to identify a broad spectrum of toxic algal species but also to quantify very low concentrations of them without filtration of large volume of water sample is still a challenge.

Chemical sensors are an attractive alternative for environmental monitoring. They offer several advantages with respect to conventional methodologies regarding robustness, versatility, portability, the possibility of integration into multisensors probes and flow systems and of course their mass production and low cost,

* Corresponding author at: Laboratoire ARAGO, Avenue Fontaulé, F-66650 Banyuls/Mer, France. Tel.: +33 46430192400; fax: +33 468881699.

E-mail address: jahir.orozco@obs-banyuls.fr (J. Orozco).

etc [16]. Biosensors are powerful tools for species detection. The study of toxic algal biodiversity and distribution is greatly facilitated by the application of rRNA probes. The target sequence from the species of interest is detected on the sensor chip via a sandwich-hybridisation [1,2], which have many advantages comparing to other methods including optical ones. The assay format maximizes discrimination of the target sequences, thus improving specificity and minimizing potential nonspecific cross reactions. Besides, it is amenable to automation, purification of rRNA is not required, reactions are rapid and easy to execute and quantification of the target species can be performed using smaller, portable and less expensive instrumentation.

The underlying principle of a sandwich hybridisation is that a species capture probe is immobilised over the electrodes. If a target sequence binds to the immobilized capture probe on the working electrode, its detection takes place via a hybridisation reaction with a second signal probe that is coupled to digoxigenin [3], which is later coupled to an antibody with a horseradish-peroxidase (HRP) enzyme. Target and signal probes bind to the capture-modified chips by incubating all the components in a hybridization mixture, after which the electrochemical detection takes place. HRP catalyses the reduction of hydrogen peroxide to water. Reduced peroxidase is regenerated by a mediator, which is reduced back at the working electrode by applying a fix potential. Once the potential is applied, a potentiostat measures the resulting electrochemical current, which can only be measured if the target nucleic acid sequence binds to both the capture and signal probes [3]. In field samples total rRNA would be extracted and this represents the target nucleic acid for the probes. For each target species, the RNA concentration per cell has to be determined. A calibration curve must be developed for each new probe set in order to determine the signal intensity at different RNA concentrations. Using the information on the curve, the electrical measurement of the potentiostat can be related to cell numbers in any field sample.

This work illustrates the electrochemical performance of a DNA-based sensor device for detecting toxic algae. Optimization of the hybridization process has being already reached in a previous work [3,4]. However, the main achievements reported herein come from elucidation of the different steps of the fabrication process from the electrochemical point of view, which will result in the improvement of the biosensor performance in terms of higher sensitivity and enhanced limits of detection when tested real samples. In this context, the effect of a mixed self-assembled monolayer (SAM) and the use of bovine serum albumin (BSA) as a blockage agent in the electrochemical signal as well as the mediator/substrate concentration ratio were studied. Proof of concept with different algal species was then investigated. Special attention was also given to the evaluation of the influence of the transducer platform geometry and material in the biosensor analytical performance.

2. Materials and methods

2.1. Apparatus and materials

Voltammetric measurements were performed with an EmStat (Electrochemical Sensor Interface), using the PSTrace software from Palm Instruments BV (The Netherlands) and a boxer connector DSC from Dropsens (Spain). Measurements were carried out with a three electrode cell printed over ceramic substrates, whose total volume is 50 μ l. This cell consists of a DNA-modified working electrode and an on-chip counter and pseudoreference electrode of different materials. 4 mm diameter screen printed gold electrodes DRP-220AT (SPAUE-4) and single wall carbon nanotubes DRP-110SWCNT (SWCNT-4) were purchased from Dropsens (Spain), 1 mm diameter screen printed gold electrodes AC1W1R1 (SPAUE-1) and graphite electrodes AC1W4R1 (SPGE-1) were from BVT tech-

Table 1

Characteristics of the screen printed gold (SPAUE-4, SPAUE-1), graphite (SPGE-1) electrodes and single wall carbon nanotubes electrodes (SWCNT-4).

Abbreviated name	Area (mm ²)	Electrode		
		Working	Counter	Reference
SPAUE-4	12.57	Au	Pt	Ag/AgCl
SWCNT-4	12.57	Carbon nanotubes	C	Ag
SPAUE-1	0.79	Au	Au	Ag/AgCl
SPGE-1	0.79	Graphite	Au	Ag/AgCl

nologies (Czech Republic). All the characteristics are summarized in Table 1. All the potentials reported refer to a Ag/AgCl pseudoreference electrode except when using SWCNT-4, which potential is refer to Ag pseudoreference electrode.

2.2. Reagents and solutions

Synthetic 18 and 38-mer oligonucleotides probes were obtained from Thermo Scientific (Germany) as powders and then diluted in 18 M Ω cm ultrapure deionized water (free of DNA and RNA) to have 100 μ M stock solutions. They are listed in Table 2. A 3.48 μ g μ l⁻¹ salmon sperm DNA solution was also used as blocking reagent in the hybridization mixture.

50 mM NaHCO₃ buffer solution (pH 9.6), 0.5 M NaH₂PO₄-H₂O buffer solution containing 1.54 M NaCl (10 $\times$ PBS, pH 7.4), 0.1 M Tris solution containing 0.3 M NaCl (Bead buffer, pH 7.6), 0.08 M Tris solution containing 0.3 M NaCl and 0.04% sodium dodecyl sulphate (SDS) (4 $\times$ hybridization buffer, pH 8.0), 0.5 M NaH₂PO₄-H₂O buffer solution containing 1.0 M NaCl (10 $\times$ POP buffer, pH 6.45), and 1.0 M PBS containing 0.1% bovine serum albumin (BSA) and 0.05% Tween 20 (PBS-BT, pH 7.4). All were prepared by diluting analytical grade reagents purchased from Sigma-Aldrich (France) in 18 M Ω cm deionized water and stored at 4 °C when not in use.

Sulphuric acid, potassium hexacyanoferrate, neutravidin, mercaptoethanol, mercaptohexanol and hydrogen peroxide were supplied by Sigma (France). N-phenyl-p-phenylenediamine monohydrochloride 99% (ADPA) was purchased from Acros Organics (Belgium) and anti-digoxigenin-HRP fragments were fabricated by Roche (France). All the other chemicals employed were of analytical reagent grade.

2.3. Experimental procedure

2.3.1. Characterization of the transducer platform

The SPAUE-4 and SPAUE-1 were pretreated by placing on their surface a 50 μ l drop of a 0.5 M H₂SO₄ solution containing 0.01 M KCl and cycling the potential between 0.0 and +1.2 V up to the voltammetric response is constant (around 30 times for the first mentioned and 15 times for the last ones) with a scan rate of 100 mV s⁻¹. Then, the electrodes were rinsed with water and dried with a vacuum pump. Their behaviour in acidic solution was analogous to that reported in literature for gold wire electrodes [17]. Thus, characteristic gold oxide formation and reduction peaks were observed at +1.15 V ($\pm$ 2.0%, n =5) and +0.45 V ($\pm$ 3.4%, n =5) for SPAUE-4 and +1.07 V ($\pm$ 3.5%, n =5) and +0.54 V ($\pm$ 3.2%, n =5) for SPAUE-1, respectively.

All the electrodes were then characterised by cyclic voltammetry in a 1 M KCl solution containing 1 mM K₃Fe(CN)₆ at a scan rate of 50 mV s⁻¹, when a potential scan ranging from -0.25 to +0.5 V was applied. Voltammograms showed a well-defined reversible oxidation and reduction peaks with a formal potential ($E_{1/2}$) of +1.20 V, i_{red} of 15.67 μ A ($\pm$ 6.4%, n =5), i_{red}/i_{ox} of 0.97 ($\pm$ 2.0%, n =5) and a peak potential difference of 0.14 V ($\pm$ 1.9%, n =5), for the SPAUE-4 and ($E_{1/2}$) of +1.22 V, i_{red} of 0.96 μ A ($\pm$ 6.1%, n =5), i_{red}/i_{ox} of 0.92

Table 2
DNA probe sets.

Abbreviated Name	Sequence (5'–3' direction)	Modifications
B-Prym	CAG CCG ACG CCG AGC GCG	5' Biotin
T-Prym	CAG CCG ACG CCG AGC GCG	5' Thiol
B-Gym	TTT TTA AAA GAT TAC CCA ATC	5' Biotin
T-Gym	TTT TTA AAA GAT TAC CCA ATC	5' Thiol
D-Prymnext	CGC CAT CCT ACC AGG CTC	3' Digoxigenin
D-Gymnext	CTG TCG GAC AAG GTC GTA	3' Digoxigenin
<i>Prymnesium parvum</i>	GAG CCT GGT AGG ATG GCG CGC GCT CGG CGT CGG CTG	
<i>Gymnodinium catenatum</i>	TAC GAC CTT GTC CGA CAG GAT TGG GTA ATC TTT TAA AAA	
Negative control	ATA TAG ATT ATC ACC AGT TGT AAG AAC CTT CCT TAA	

($\pm 2.6\%$, $n = 5$) and a peak potential difference of 0.95 V ($\pm 6.1\%$, $n = 5$), for the SPAuE-1. In both cases the roughness factor was around 1.1 and the surface area increased less than 20%. Similar results were obtained when characterized a set of 5 SWCNT and 5 SPG-1, respectively. Whole results show the excellent performance and high response reproducibility attained in all the cases.

2.3.2. Development of the biosensor device

The target sequence from the species of interest is detected on the sensor chip via a sandwich-hybridisation [18,19], (Fig. 1). The underlying principle of the sandwich hybridisation is that a capture specific probe is either directly immobilized over the SPAuE-4 and SPAuE-1 electrodes by an overnight incubation in $10 \mu\text{M}$ T-Prym probe or via a 30 min incubation in $10 \mu\text{M}$ B-Prym probe in the case of the SWCNT-4 and SPGE-1 electrodes. In the latter case, the attachment of the biotinylated probe is performed through Neutravidin, for which the chips were previously incubated overnight in 0.5 mg ml^{-1} Neutravidin solution. Concentration of Neutravidin was previously optimized for both carbon surfaces (SWCNT-4 and SPGE-1, results not shown). Concentrations that lead to better positive to negative control ratio were used in this approach as optimal concentrations. If a target sequence is bound to the immobilized capture probe on the working electrode, its detection takes place via a hybridisation reaction to a second signal probe that is coupled to digoxigenin (D-Prymnext) [14], which is later coupled to an antibody with a horseradish-peroxidase (HRP) enzyme. Probes were designed with the probe match function within ARB software [20] with mismatches placed as close to the center of the probe as possible to destabilize the helix formation with non-target RNA. Only one of the two probes has to be specific for the target species. Target and signal probes are assembled over the capture-modified chips in a unique step by incubation in a hybridization mixture contain-

ing $1 \mu\text{l}$ Dig-marked DNA probe ($1.4 \mu\text{M}$), $3.5 \mu\text{l}$ $4\times$ hybridization buffer, $1 \mu\text{l}$ salmon sperm DNA and $7.5 \mu\text{l}$ milliQ water and either $1 \mu\text{l}$ target probe (36 bases, $1.4 \mu\text{M}$) for the positive control and $1 \mu\text{l}$ non complementary DNA for the negative control. Chips were incubated in the hybridization mixture at 46°C during 30 min, after which the electrochemical detection takes place. It is necessary to remark that the effect of capture probe density on signal level and hybridization efficiency was empirically investigated (data not shown) and optimal conditions used here.

HRP catalyses the reduction of hydrogen peroxide to water. Reduced peroxidase is regenerated by p-aminodiphenylamin (ADPA), which functions as a mediator. The oxidised mediator is reduced back at the working electrode with a potential of -0.15 V taking an electron from the surface of the sensor (Fig. 1). The electrochemical cell of $50 \mu\text{l}$ contains of 0.2 mM ADPA, 10 mM H_2O_2 in the POP buffer background. Once the potential is applied between the working and the reference electrode, the potentiostat measures the resulting electrochemical current, activated through the flow of the electrons from the surface, which can only be measured if the target nucleic acid sequence binds to both the capture and signal probes and thus must be present in the sample to be analyzed [14]. For each target species, the RNA concentration per cell has to be determined. A calibration curve must be developed for each new probe set developed in order to determine the signal intensity at different RNA concentrations. Using the information on the curve, the electrical measurement of the potentiostat can be related to cell numbers in a field sample.

3. Results and discussion

3.1. Voltamperometric characterisation

First experiments were conducted using synthetic 18-mer oligonucleotides probes from *Prymnesium parvum*. The biosensor performance is based on the catalytic reduction of H_2O_2 to H_2O , which takes place with the concomitant oxidation of p-aminodiphenylamine (ADPA). These oxidized species could be reduced back by polarizing the electrode. Results are presented in Fig. 2. Voltammogram in POP background showed that no direct electron transfer between the heme group of the HRP structure and the electrode took place (1). Signal was also recorded in POP containing 0.2 mM of ADPA, where anodic and cathodic faradaic currents from the ADPA electrochemical process were recorded (2). In the presence of H_2O_2 , a large cathodic current was recorded, which is related to the reduction of the oxidized ADPA species generated by the HRP catalytic process (3 and 4). A similar study was performed using methylene blue (MB) as mediator (results not shown). However the considerable higher current densities achieved when ADPA mediated the reaction each case proves its better performance as compared to MB. In consequence ADPA and a set potential of -0.15 V vs pseudo Ag/AgCl was chosen for the next experiment.

Analytical signal was studied also by amperometry. This was first performed with no optimized chemical conditions (0.2 mM

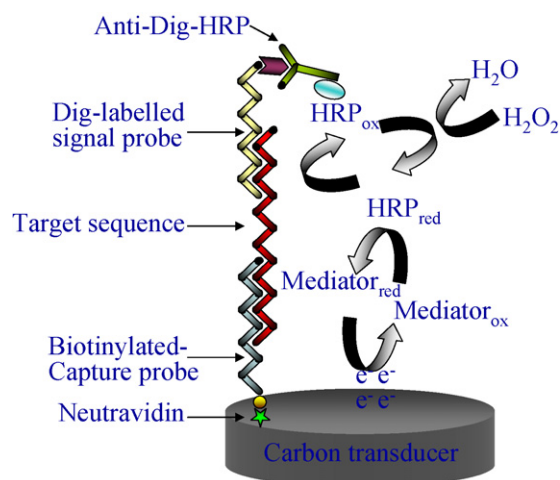


Fig. 1. Principle of sandwich hybridisation of a target probe between a capture and a signal probe.

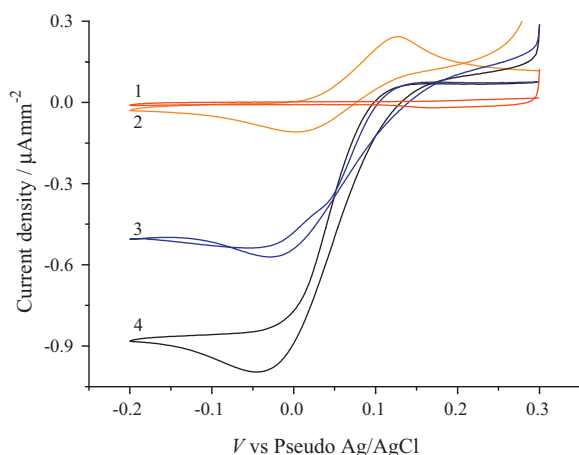


Fig. 2. Biosensor catalytic responses by cyclic voltammetry, at a scan rate of 0.01 V s^{-1} . Signals were recorded in: (1) POP, (2) POP containing 0.2 mM of ADPA, (3) POP containing 0.2 mM ADPA and 5 mM of H_2O_2 , (4) POP containing 0.2 mM ADPA and 10 mM H_2O_2 .

ADPA as mediator and 5.0 mM H_2O_2 as substrate). Results are depicted in Fig. 3 comparing SWCNT-4 and SPGE-1, but similar results were obtained when compared SPAuE-4 and SPAuE-1. A fix potential of -0.15 V was applied, in the presence of H_2O_2 the current was registered during 120 s . As shown in Fig. 3 (3) the maximum current density value ($1.48 \mu\text{A mm}^{-2}$) was attained quickly, but a decrease of this value was also observed with time. This falling current might be attributed to depletion of ADPA at the electrode surface. ADPA is supposed to be replenished by diffusion from bulk solution. However, the flux in absence of stirring is not enough to replace the analyte as fast as it is consumed. To circumvent this drawback a couple of strategies, such as the employment of a microelectrode, with smaller dimensions, where radial diffusion of substrate is radial rather than linear [16], or the use of a microfluidic system that guarantees the constant flow of analyte to the surface of the electrode could be implemented. Herein, although no microelectrode was tested (it is not the objective of this work) the effect of diminishing the electrode diameter up to 1 mm using identical immobilization, hybridisation and detection conditions was evaluated. Fig. 3 (4) shows the almost 7-fold highest current density achieved with the SPGE-1 mm device ($10.09 \mu\text{A mm}^{-2}$). Comparable results were also obtained when using the SPAuE-1 chip. The diffusion flux of substrate towards the electrode surface turned out

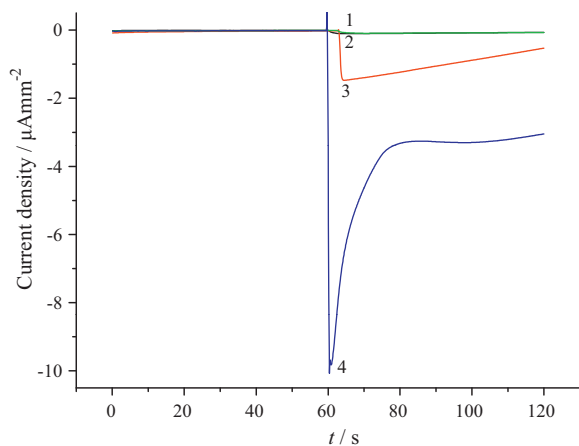


Fig. 3. Analytical signal recorded at a fix potential of -0.15 V . (1) POP background, (2) negative control, (3) positive control using a SWCNT-4 mm electrode, (4) positive control using the SPGE-1.

to be higher than expected, with a higher decrease in the signal at the beginning (during the first 15 s after the electrochemical reaction) and almost a plateau after this time was observed. This plateau indicates the slow decrease in response as ADPA is depleted. As all the conditions were kept constant, and only diameter was varied, the improved response of the biosensor ascribed to the smallest dimensions of the electrode. Better results would be expected by diminishing the electrode up to micrometric dimensions, i.e., by means of using microelectrode arrays [16]. On the other hand, in a stirred solution, the local hydrodynamics are such that the flux of ADPA to the surface is higher and no variations are expected with time, thus attaining a steady-state current. Because stirring is not possible with this experimental set up, the estimated maximum current density value was used for all the calculations in this work. However work, exploiting the optimized DNA-based device integrated in a flow system, is ongoing.

3.2. Optimization of the fabrication process

The effect of a mixed self-assembled monolayer (SAM) and the use of bovine serum albumin (BSA) as a blocking agent in the electrochemical signal was then studied. Preparation of mixed monolayers formed by a thiol-thethered DNA probe and a spacer thiol has been reported [21,22]. It was demonstrated that after formation of the DNA-thiol SAM a post-treatment with mercapto-1-hexanol lead to thiol-tethered oligonucleotides being lifted off the surface, so that the S atom is the only anchoring point [23]. On the other hand, in the case of SWCNT-4 electrodes in spite of the highly non specific adsorption occurred, the use of neutravidin(avidin)-enzyme conjugates has been widely carried out to anchorage of biotinylated DNA and RNA [24]. Likewise, BSA has been effectively tested as blocking agent to minimize this undesirable adsorption [25]. Herein, the use of 1 mM mercapto-ethanol as spacer in the SPAuE-4 platform and BSA as blockage agent in both SPAuE-4 and SWCNT-4 transducers and its influence in the analytical signal was interrogated. In the first case, results presented in Fig. 4 show the synergetic effect of the mixed DNA-monolayer and BSA, which increases almost twice ($0.80 \mu\text{A mm}^{-2}$) the current density respect to a biosensor made without these added ($0.48 \mu\text{A mm}^{-2}$). However, each treatment by itself leads to worse results (0.11 and $0.21 \mu\text{A mm}^{-2}$), respectively. Biosensors fabricated on SWCNT-4 platforms – using neutravidin(avidin)-enzyme conjugates – without BSA as blockage agent lead to poorer results as expected (date not shown).

Some authors have reported the reproducible covalent immobilization of thiol-tethered oligonucleotides achieved without any pre-treatment of the gold surface [25]. In contrast, as it is well known, an exhaustively cleaned gold surface is required for the formation of SAMs of thiols [26,27]. In this context, effect of the pre-treatment the SPAuE-4 previous immobilization of the thiol-RNA capture probe was studied. This pre-treatment consists of either cleaning the chips with piranha solution (chemical pre-treatment) or cycling 30 times in a 0.1 M $\text{H}_2\text{SO}_4/0.1 \text{ M}$ KCl solution (electrochemical pre-treatment) or both (worse results, not shown). The best results were obtained with the electrochemically activated electrodes ($1.39 \mu\text{A mm}^{-2}$), where the signal improved 1.6 times compared to the non-pre-treated ones ($0.48 \mu\text{A mm}^{-2}$). From results summarized in Fig. 4 is evident that the gold activated surface, the thiol posttreatment and BSA blockage have a remarkable positive effect over the intensity of the amperometric signal. On the other hand, in the case of SWCNT-4 blockage of the surface with BSA effectively avoided unspecific adsorption. Both practices are implemented in the biosensor fabrication protocol for the next experiments.

Coming back to the SAMs formed on a metal surface, it is well known they depends not only on the chemical nature of the thiol

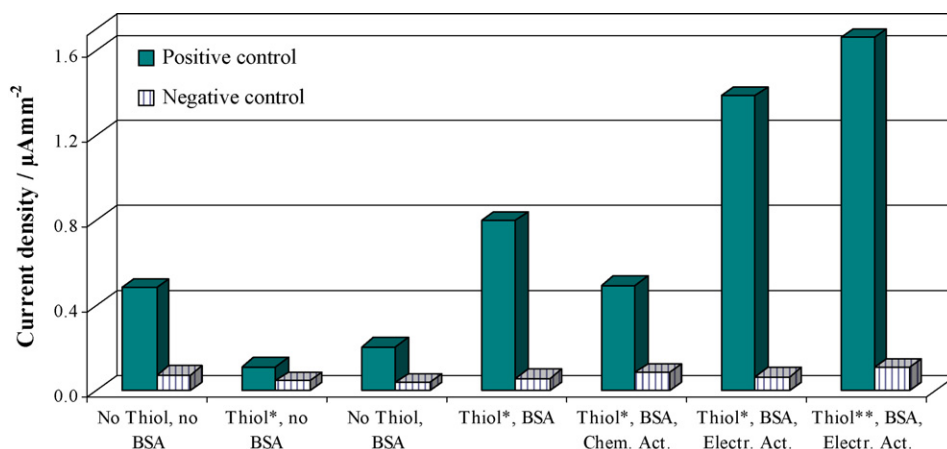


Fig. 4. Effect of formation of a mixed monolayer, use of BSA as blockage agent and a pre-treatment of the SPAuE-4 in the electrochemical signal reached. Signals were recorded in POP containing 0.2 mM ADPA and 5 mM H₂O₂. r.s.d. lower than 5.6% each case ($n=3$). *Mercaptoethanol, **Mercaptohexanol.

used but also on their length [27]. In this step the effect of mercaptoethanol (a shorter chain thiol) in the analytical signal was compared with mercaptohexanol (media length thiol). Mercaptoethanol seems to straighten better the thiol-probes, thus allowing a better exposure to the target sequence, producing higher signals ($1.66 \mu\text{A mm}^{-2}$) and improving reproducibility among devices (s.d 0.05 $n=4$). Based on that mercaptohexanol was chosen for further experiments. Comparative results are presented in Fig. 4.

3.3. Optimization of the concentration of ADPA/substrate and biosensor analytical performance

Digoxigenin specific antibody coupled to HRP, the enzyme-linked immunomarker, has been used in this study to follow the electroanalytical signal as mentioned. O-phenylenediamine (ODP) and 2,2 bis(3-ethylbenzenediamine-6-sulphonic acid) (ABTS) are the traditional mediators in the reductive catalysis of H₂O₂ to H₂O. However, they have the drawback of being mutagenic in the Ames test [28,29]. In order to overcome this dilemma, p-aminodiphenylamine (ADPA), neither a mutagenic nor carcinogenic mediator of this reaction, was tested in this work. A study of different concentrations of mediator/substrate was performed by amperometry, using SWCNT-4 electrodes at the electrochemical conditions mentioned. The best signal for the positive/negative control ($2.15/0.21 \mu\text{A mm}^{-2}$) was obtained with 0.4 mM ADPA and 10 mM H₂O₂ as shown in Fig. 5, which is in agreement with results previously obtained [15].

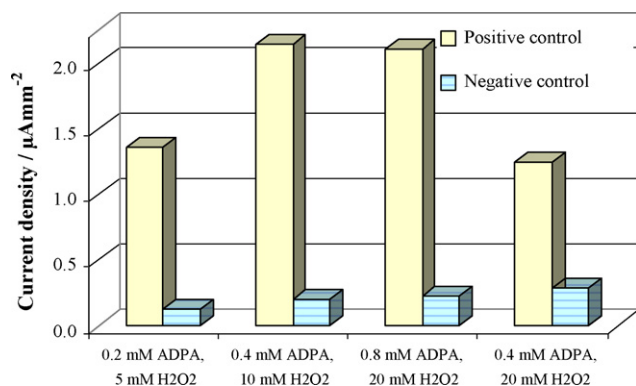


Fig. 5. Optimization of the concentration of mediator and substrate. r.s.d lower than 7.2% each case, ($n=3$).

Another test was performed by combining the effect of the optimized fabrication process (Section 3.2), the better mediator/substrate concentration ratio obtained above and the smaller electrode (SPGE-1). The improved response, $9.98 \mu\text{A mm}^{-2}$, s.d 0.3, $n=4$ for the positive control and $0.14 \mu\text{A mm}^{-2}$, s.d 0.08, $n=4$ for the negative control, turned out to be almost a 5-fold increase and 1.5-fold decrease, respectively, as compared to the respective signals obtained with the SWCNT-4 electrodes under optimal conditions. On the other hand, comparable results were also obtained when using SPAuE-1, which reveals that both materials (gold and graphite) have suitable conductive features for being used as transducer platform in this approach. A thiolated capture probe can be directly immobilized over the gold surface, thus avoiding the overnight neutravidin incubation step necessary when using the graphite chips. However, gold chips need an additional electrochemical activation step as showed above.

The working response and performance characteristics of the DNA-based sensor were evaluated by amperometry. Fig. 6 shows the quiescent response of the device when different concentrations of target probe, up to $26.9 \text{ ng } \mu\text{l}^{-1}$, were set to react with the capture and signal probes. The decrease of the signal at this last concentration (dotted line), suggests that the limit of concentra-

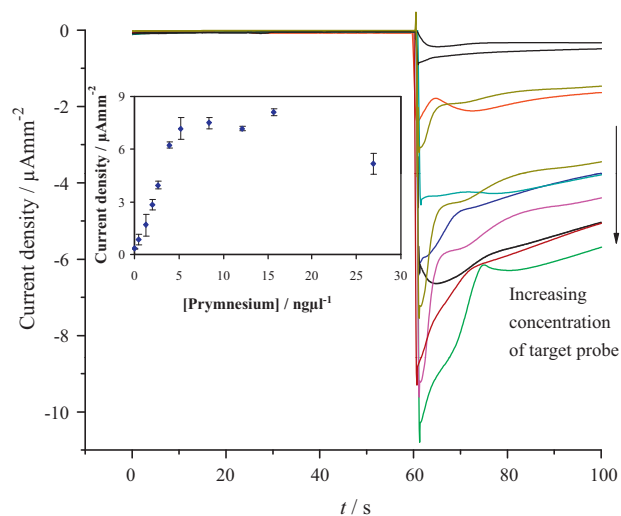


Fig. 6. Biosensor amperometric response recorded when different concentrations of target probe, up to $26.9 \text{ ng } \mu\text{l}^{-1}$, were set to react with the capture and signal probes. Inset, the corresponding calibration curve.

tion in which the maximum quantity of capture, target and signal probe strands are linked was overcome. Inset, the corresponding calibration curve, which resulted from plotting the estimated current density value obtained from each concentration, against the concentration of the target probe (bar errors in each point indicates the standard deviation of four measures). Standard deviations among results (low than 8% each point) demonstrated the high reproducibility among different devices. From the calibration curve the linear range was estimated, being that ranged from 0.0 to $5.2 \text{ ng } \mu\text{L}^{-1}$, with a sensitivity (slope) of $1.41 \text{ } \mu\text{A } \mu\text{L ng}^{-1} \text{ mm}^{-2}$, an intercept of $0.16 \text{ } \mu\text{A mm}^{-2}$ and a linear regression coefficient of 0.98, $n=26$. The limit of detection (LOD) was estimated using the $3S_B/m$ IUPAC criteria, where m is the slope of the linear calibration plot and S_B is the standard deviation of the amperometric responses of the negative control reading. The LOD obtained was $0.058 \text{ ng } \mu\text{L}^{-1}$ of synthetic *Prymnesium parvum*. A calibration curve using RNA extracted from this algal species has to be performed when field samples are analyzed.

3.4. Proof of concept with other species

To evaluate the feasibility of target sequence using the biosensor platform developed for sensing others algal species, the protocol was interrogated using the *Gymnodinium catenatum* 18S with the capture, target and signal probes arranged over SWCNT-4 transducer platforms in the same orientation as in the previous tests (*Prymnesium parvum* experiments). This arrangement is such that the strand biotinylated extreme is distal to the Dig-HRP mark. The signals for the *Gymnodinium catenatum* 18S at optimal conditions were higher with respect to the initial ones as in *Prymnesium parvum*. On the other hand, this optimized signal ($0.7 \text{ } \mu\text{A mm}^{-2}$) was lower compared to that obtained from *Prymnesium parvum* ($2.15 \text{ } \mu\text{A mm}^{-2}$), under the same conditions. On the contrary the signal for the negative control was higher ($0.28 \text{ } \mu\text{A mm}^{-2}$) for the *Gymnodinium catenatum* compared to that for *Prymnesium parvum* ($0.21 \text{ } \mu\text{A mm}^{-2}$), performed under the same conditions. Based on the aforementioned results, we can say that the biosensor platform together with the entire protocol is useful not only for sensing *Prymnesium parvum* but also other species, as e.g., *Gymnodinium catenatum*, when using the same geometric arrangement of the capture, target and signal probes (the strand biotinylated extreme distal to the Dig-HRP mark). However, our results also suggest that the range of measurement for this species is lower because of the increased negative control signal and the decreased positive control signal.

4. Conclusions

Elucidation, from the electrochemical point of view, of the different steps in the fabrication process of a DNA-based sensor device based on a sandwich type hybridization was reported here. Results revealed that the gold activated surface pre-treatment, the post-treatment with a thiol spacer and later BSA blockage have a remarkable positive effect on the intensity of the amperometric signal. The best mediator/substrate concentration was 0.4 mM ADPA and $10 \text{ mM H}_2\text{O}_2$ and the best orientation of probes vis-à-vis the target sequence is such that the capture probe with the biotinylated label should be distal to the Dig-HRP label ($5'$ – $3'$ orientation of the probe). The response of the developed platform combining the whole optimized conditions and the effect of decreasing the electrode diameter led to almost a 5-fold improved current density of the synthetic *Prymnesium parvum* positive control, which will result in the improvement of the biosensor performance in terms of higher sensitivity and enhanced limits of detection when testing real samples. Results turned out to be independent of the electrochemical material if gold electrodes are previously electrochem-

ically activated. Additionally, the biosensor concept proved with other algal species demonstrated the feasibility of using this format for identification and quantification of different toxic algal species.

References

- [1] A. Sournia, M.J. Chretiennotdinet, M. Ricard, Marine-phytoplankton – how many species in the world ocean, *Journal of Plankton Research* 13 (1991) 1093–1099.
- [2] M. Maso, E. Garcés, Harmful microalgae blooms (HAB); problematic and conditions that induce them, *Marine Pollution Bulletin* 53 (2006) 620–630.
- [3] A. Penna, M. Magnani, Identification of *Alexandrium* (Dinophyceae) species using PCR and rDNA-targeted probes, *Journal of Phycology* 35 (1999) 615–621.
- [4] C.A. Scholin, R. Marin, P.E. Miller, G.J. Doucette, C.L. Powell, P. Haydock, J. Howard, J. Ray, DNA probes and a receptor-binding assay for detection of *Pseudo-nitzschia* (Bacillariophyceae) species and domoic acid activity in cultured and natural samples, *Journal of Phycology* 35 (1999) 1356–1367.
- [5] K. Ayers, L.L. Rhodes, J. Tyrrell, M. Gladstone, C. Scholin, International accreditation of sandwich hybridisation assay format DNA probes for micro-algae, *New Zealand Journal of Marine and Freshwater Research* 39 (2005) 1225–1231.
- [6] S. Diercks, C. Gescher, K. Metfies, L.K. Medlin, Evaluation of locked nucleic acids for signal enhancement of oligonucleotide probes for microalgae immobilised on solid surfaces, *Journal of Applied Phycology* 21 (2009) 657–668.
- [7] S. Diercks, L.K. Medlin, K. Metfies, Colorimetric detection of the toxic dinoflagellate *Alexandrium minutum* using sandwich hybridization in a microtiter plate assay, *Harmful Algae* 7 (2008) 137–145.
- [8] C. Gescher, K. Metfies, L.K. Medlin, The ALEX CHIP – development of a DNA chip for identification and monitoring of *Alexandrium*, *Harmful Algae* 7 (2008) 485–494.
- [9] D.I. Greenfield, R. Marin, S. Jensen, E. Massion, B. Roman, J. Feldman, C.A. Scholin, Application of environmental sample processor (ESP) methodology for quantifying *Pseudo-nitzschia australis* using ribosomal RNA-targeted probes in sandwich and fluorescent in situ hybridization formats, *Limnology and Oceanography—Methods* 4 (2006) 426–435.
- [10] A.J. Haywood, C.A. Scholin, R. Marin, K.A. Steidinger, C. Heil, J. Ray, Molecular detection of the brevetoxin-producing dinoflagellate *Karenia brevis* and closely related species using rRNA-targeted probes and a semiautomated sandwich hybridization assay, *Journal of Phycology* 43 (2007) 1271–1286.
- [11] C. O'Halloran, M.W. Silver, T.R. Holman, C.A. Scholin, *Heterosigma akashiwo* in central California waters, *Harmful Algae* 5 (2006) 124–132.
- [12] J.V. Tyrrell, L.B. Connell, C.A. Scholin, Monitoring for *Heterosigma akashiwo* using a sandwich hybridization assay, *Harmful Algae* 1 (2002) 205–214.
- [13] D.M. Anderson, D. Kulis, D. Erdner, S. Ahn, D. Walt, Fibre optic microarrays for the detection and enumeration of harmful algal bloom species, *African Journal of Marine Science* 28 (2006) 231–235.
- [14] K. Metfies, S. Huljic, M. Lange, L.K. Medlin, Electrochemical detection of the toxic dinoflagellate *Alexandrium ostensefeldii* with a DNA-biosensor, *Biosensors & Bioelectronics* 20 (2005) 1349–1357.
- [15] S. Diercks, K. Metfies, L.K. Medlin, Development and adaptation of a multiprobe biosensor for the use in a semi-automated device for the detection of toxic algae, *Biosensors & Bioelectronics* 23 (2008) 1527–1533.
- [16] J. Orozco, C. Fernandez-Sanchez, C. Jimenez-Jorquera, Ultramicroelectrode array based sensors: a promising analytical tool for environmental monitoring, *Sensors* 10 (2010) 475–490.
- [17] Q. Wang, N.Q. Li, Y.H. Wu, Direct electrochemistry of cytochrome c at a metallothioneins self-assembled gold electrode, *Electroanalysis* 13 (2001) 149–154.
- [18] Rautio F.B.K.B., J. Lahdenpera, A. Breinstein, S. Molin, P. Neubauer, Sandwich hybridisation assay for quantitative detection of yeast RNAs in crude cell lysates, *Microbial Cell Factories* 2 (2003) 4.
- [19] N. Zammattio, P. Moris, I. Alexandre, D. Vaira, J. Piette, J. Remacle, DNA-probe hybridization in microwells using a new bioluminescent system for the detection of PCR-amplified HIV-1 proviral DNA, *Journal of Virological Methods* 55 (1995) 185–197.
- [20] S.O. Ludwig, W. Westram, R. Richter, L. Meier, H. Yadukumar, A. Buchner, T. Lai, S. Steppi, G. Jobb, W. Förster, I. Brettske, S. Gerber, A.W. Ginhart, O. Gross, S. Grumann, S. Hermann, R. Jost, A. König, T. Liss, R. Lüßmann, M. May, B. Nonhoff, B. Reichel, R. Strehlow, A. Stamatakis, N. Stuckmann, A. Vilbig, M. Lenke, T. Ludwig, A. Bode, K.-H. Schleifer, ARB, a software environment for sequence data, *Nucleic Acids Research* 32 (2004) 1363–1371.
- [21] T.M. Herne, M.J. Tarlov, Characterization of DNA probes immobilized on gold surfaces, *Journal of the American Chemical Society* 119 (1997) 8916–8920.
- [22] J. Zhang, R.J. Lao, S.P. Song, Z.Y. Yan, C.H. Fan, Design of an oligonucleotide-incorporated nonfouling surface and its application in electrochemical DNA sensors for highly sensitive and sequence-specific detection of target DNA, *Analytical Chemistry* 80 (2008) 9029–9033.
- [23] R. Levicky, T.M. Herne, M.J. Tarlov, S.K. Satija, Using self-assembly to control the structure of DNA monolayers on gold: a neutron reflectivity study, *Journal of the American Chemical Society* 120 (1998) 9787–9792.
- [24] F. Lucarelli, G. Marrazza, A.P.F. Turner, M. Mascini, Carbon and gold electrodes as electrochemical transducers for DNA hybridisation sensors, *Biosensors & Bioelectronics* 19 (2004) 515–530.
- [25] G. Carpinì, F. Lucarelli, G. Marrazza, M. Mascini, Oligonucleotide-modified screen-printed gold electrodes for enzyme-amplified sensing of nucleic acids, *Biosensors & Bioelectronics* 20 (2004) 167–175.

- [26] D.J. Kim, R. Pitchimani, D.E. Snow, L.J. Hope-Weeks, A simple method for the removal of thiols on gold surfaces using an $\text{NH}_4\text{OH}-\text{H}_2\text{O}_2-\text{H}_2\text{O}$ solution, *Scanning* 30 (2008) 118–122.
- [27] J. Orozco, C. Jimenez-Jorquera, C. Fernandez-Sanchez, Gold nanoparticle-modified ultramicroelectrode arrays for biosensing: a comparative assessment, *Bioelectrochemistry* 75 (2009) 176–181.
- [28] C.E. Voogd, J.J. Vanderstel, J. Jacobs, On the mutagenic action of some enzyme-immunoassay substrates, *Journal of Immunological Methods* 36 (1980) 55–61.
- [29] E.S. Bos, A.A. van der Doelen, N. van Rooy, A.H.W.M. Schuurs, 3,3', 5'-Tetramethylbenzidine as an Ames test negative chromogen for horse-radish peroxidase in enzyme-immunoassay, *Journal of Immunoassay* 2 (1981) 187–204.

Biographies

Jahir Orozco has received his B.Sc. in Chemistry from University of Antioquia (Medellín, Colombia) and his PhD in Chemistry from University of Barcelona

and Institute for Microelectronics of Barcelona (Barcelona, Spain), in 2008. His research interests are in Electrochemistry and Environmental Analytical Chemistry, particularly in the development and application of chemical (bio)sensors and (micro)nanodevices, integration of nanostructured/bio materials with electrochemical transducers and modification of surfaces and interfaces for selective recognition. Currently he is a postdoc in the University of Pierre and Marie Curie Paris at Laboratoire Océanographie Microbienne, Observatoire Océanologique de Banyuls sur Mer, France developing DNA-based sensors for detecting toxic algae and fresh water pathogens.

Linda K. Medlin works at the University of Pierre and Marie Curie Paris at the Observatoire Océanologique de Banyuls-sur-mer, France. She obtained her Ph.D. from Texas A&M University, USA. Professor Medlin's research interests are in marine phytoplankton evolution, and she has produced phylogenies, revised taxonomies and molecular clocks for the diatoms, dinoflagellates and haptophytes. Recent research efforts involve species specific probes for early warning systems for toxic algae and for phylochips for biodiversity analysis using microarrays. She has 194 research papers including 35 book chapters. She has participated in 16 EU grants and many UK and German national projects.